



Assignment-3

**Web Programming using
Python and JavaScript**

Submitted By:

Name: Aditya Pal

Course: B.Tech CSE

Year: 2nd Year, 4th Semester

Roll No: 2401010119

Submitted To:

Name: Dr. Mansi Kajal

❖ **AIM:** To design and develop a full-stack quiz web application using HTML, CSS, Python, and Flask.

❖ **Learning Objectives:**

- Create structured web pages using HTML
- Apply styling using CSS
- Build backend using Flask
- Process form data using Python
- Integrate frontend and backend

❖ **Problem Statement:** Develop a web-based Quiz Master Application with at least 5 MCQs, submission, and score display.

❖ **Project Structure:** quiz_master/ |— app.py |— templates/
| |— index.html | |— quiz.html | |— result.html |—
static/ |— style.css

❖ **Functional Requirements:**

- Home Page with Start Quiz button
- Quiz Page with MCQs
- Flask Backend for routing and logic
- Result Page showing score and feedback
- CSS Styling for UI

❖ **Tasks:**

- Setup Flask app
- Create quiz data

- Design HTML templates
- Process form submission
- Apply CSS styling

❖ **Mandatory Features:**

- Flask working app
- Minimum 5 questions
- Proper UI pages
- Score calculation
- Feedback message

❖ **Bonus Features:**

- Restart quiz
- Timer
- Negative marking
- Shuffle questions

❖ **CODES:**

➤ **PYTHON Code(app.py):**

```
from flask import Flask, render_template, request, redirect, url_for
import random

app = Flask(__name__)

# Quiz Data
quiz_data = [
    {
        "question": "What is the capital of India?",
        "options": ["Mumbai", "Delhi", "Kolkata", "Chennai"],
        "answer": "Delhi"
    },
    {
        "question": "Which language is used for web apps?",
```

```

        "options": ["Python", "JavaScript", "Java", "All of the above"],
        "answer": "All of the above"
    },
    {
        "question": "Who developed Python?",
        "options": ["Dennis Ritchie", "Guido van Rossum", "James Gosling",
"Bjarne Stroustrup"],
        "answer": "Guido van Rossum"
    },
    {
        "question": "What does HTML stand for?",
        "options": ["Hyper Trainer Marking Language", "Hyper Text Markup
Language", "Hyper Text Marketing Language", "None"],
        "answer": "Hyper Text Markup Language"
    },
    {
        "question": "Which CSS property controls text size?",
        "options": ["font-style", "text-size", "font-size", "text-style"],
        "answer": "font-size"
    }
}
]

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/quiz')
def quiz():
    random.shuffle(quiz_data) # Bonus: shuffle
    return render_template('quiz.html', quiz=quiz_data)

@app.route('/result', methods=['POST'])
def result():
    score = 0
    total = len(quiz_data)

    for i in range(total):
        selected = request.form.get(f"q{i}")
        if selected == quiz_data[i]['answer']:
            score += 1

    if score == total:
        feedback = "Excellent! 🏆"
    elif score >= 3:
        feedback = "Good Job! 👍"
    else:
        feedback = "Try Again! 💡"

```

```

    return render_template('result.html', score=score, total=total,
feedback=feedback)

if __name__ == '__main__':
    app.run(debug=True)

```

➤ HTML Code(index.html):

```

<!DOCTYPE html>
<html>
<head>
    <title>Quiz Master</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css')
    }}">
</head>
<body>
    <div class="container">
        <h1>Welcome to Quiz Master 🎯</h1>
        <a href="/quiz"><button>Start Quiz</button></a>
    </div>
</body>
</html>

```

➤ HTML Code(quiz.html):

```

<!DOCTYPE html>
<html>
<head>
    <title>Quiz</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css')
    }}">
</head>
<body>
<div class="container">
    <h2>Quiz Questions</h2>

    <form action="/result" method="POST">
        {% for q in quiz %}
            <div class="question">
                <p>{{ loop.index }}. {{ q.question }}</p>
                {% for option in q.options %}
                    <input type="radio" name="q{{ loop.index0 }}" value="{{
option }}" required>
                    {{ option }}<br>
                {% endfor %}
            </div>
        {% endfor %}
        <button type="submit">Submit</button>
    </form>
</div>

```

```
    </form>
</div>
</body>
</html>
```

➤ HTML Code(result.html):

```
<!DOCTYPE html>
<html>
<head>
  <title>Result</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css')
  }}">
</head>
<body>
<div class="container">
  <h1>Your Score: {{ score }} / {{ total }}</h1>
  <h2>{{ feedback }}</h2>

  <a href="/quiz"><button>Restart Quiz</button></a>
</div>
</body>
</html>
```

➤ CSS Code(style.css):

```
body {
  font-family: Arial, sans-serif;
  background: linear-gradient(to right, #667eea, #764ba2);
  color: white;
  text-align: center;
}

.container {
  margin-top: 50px;
  padding: 20px;
}

button {
  padding: 10px 20px;
  font-size: 16px;
  border: none;
  border-radius: 5px;
  background-color: #ffcc00;
  cursor: pointer;
}

button:hover {
```

```
background-color: #ffaa00;
}

.question {
  margin: 20px 0;
  text-align: left;
  display: inline-block;
}
```

❖ OUTPUT:

